

Supervised Learning - Report

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1 Introduction

Template introduction paragraph: "Given [features] describing [each observation], we aim to predict [label] for new, unseen [observations], and to determine which model and hyperparameters generalize best, compared with [benchmark]."

Describe the problem and the data. Explain what you are trying to accomplish and what is the status quo and/or benchmark for this problem.

2 Process

Describe the process for finding the best model and justifying all decision. If you used other papers/sources for your decisions you need to reference them here.

After having picked the dataset, we needed to verify it's quality and usability for a potential model. We conducted this part of the process by performing simple, but effective EDA which involved

2.1 Data Cleaning

The initial steps to was to clean the data. **The _return Columns** The columns red_return and blue_return are columns who's values reflect the payout a \$ 1 bet would return. Including them would cause target leakage since they tell the model exactly who won (loser's return being 0), causing a 100% model prediction rate accuracy.

Stances

We noticed in the red_stance and blue_stance columns that there were a few rows with faulty classifications ("Southpaw " with a trailing space and

"Unknown") that needed to be excluded. We also found one other classification called "Open Stance", which describes an opposite stance matchup, which was only found in the blue_stance column. Excluding it made sense on the premise that it conflicts with our own stance classifications.

Catch Weight

In the column weight_class there is a value called "Catch Weight" which describes fights where one or both fighters missed their target weights and therefore fought outside one of the 13 regulatory weight classes. The reason for dropping this is because:

1. It's an irrelevant outlier that might conflict with the standard classification we're planning to use
 2. It only consists of 72 rows out of 6000+
- _diff Column Outliers

In the rounds_diff column we found one extreme outlier where the value was 440. We excluded this due to it being an impossible data value when looking at either fighters records (i.e. total fights).

In the reach_diff column there is a single outlier where the reach difference was 186 caused by one of the fighters reach being measured 0 cm. This is an obvious error caused by a missing data value so we naturally excluded it.

3 Results

Report on the performance for the different models and hyperparameters. Essentially, take the numbers you got as output of your program and put them into a nice form (plots, tables, ...) that makes them easier to interpret and shows, for example, how the choice of values for different hyperparameters influences the performance.

4 Conclusion

Interpret the results and compare with results in the literature (if possible). What can be said about the performance of different models on the problem? How important is the choice of the hyperparameters for the performance of the different models?

5 Future Work

Suggest how your results could be improved. What did you learn from the results you got and what would you do differently now that you have learned

this?